

**CS6P008 - Deep Learning Lab**  
Lab 7, Date: March 23, 2026  
**Topic(s): Convolutional Neural Networks**

Timing: 10:00 AM to 01:00 PM

Max Points: 10

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**Instructions**

1. Provide commented, indented code. Variables should have meaningful names.
2. Submit two .ipynb files.  
The names should be `[student name]_lab[number]_1.ipynb` (before completion of the lab at 1 PM) and `[student name]_lab[number]_2.ipynb` (your final code file by the deadline).
3. Read the questions carefully before answering. If a question asks to follow a particular approach or to use a specific data structure, then it must be followed.
4. Write questions in separate text blocks in Jupyter/Colab Notebook before the code blocks containing answers.
5. All plots should have appropriate axis labels, titles, and legends.

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**Use only deep learning modules (PyTorch / TensorFlow). Do NOT implement convolution or pooling from scratch.**

1. Design and implement a Convolutional Neural Network (CNN) using predefined modules (not from-scratch implementation) and perform the following tasks:
  - Load the STL-10 dataset (available in `torchvision` and `tensorflow_datasets`) and perform necessary pre-processing steps such as normalization and standardization.
  - Design a Convolutional Neural Network (CNN) architecture consisting of 7 layers, including 5 Convolutional layers and 2 Fully Connected layers. Use the modules, such as `Conv2D`, `ReLU`, `Max/Avg Pooling`, and `Fully Connected` layers.

**INPUT**  $\rightarrow$   $[[\text{CONV} \rightarrow \text{RELU}] * N \rightarrow \text{POOL?}] * M \rightarrow [\text{FC} \rightarrow \text{RELU}] * K \rightarrow \text{FC}$

The architecture should satisfy the following conditions:

- The number of feature maps (channels) in the convolutional layers should be 32, 64, 128, 256, and 512, respectively.
- The spatial dimensions (height and width) should be reduced by a factor of 2 using pooling layers after appropriate convolutional layers.
- Use appropriate kernel size ( $3 \times 3$  or  $5 \times 5$ ) and padding wherever necessary to preserve spatial dimensions.
- Train the model until convergence using an appropriate optimizer, loss function, batch size, weight initialization, learning rate, learning rate scheduler, and early stopping criteria.

- Evaluate the trained model on the test dataset and report the accuracy.
  - Plot training loss and accuracy curves.
  - **(Optional)** Modify the network by changing the kernel size as well as the number of channels and observe the effect on performance.
2. Answer the following questions based on your implementation and experimental observations:
- Modify your CNN by using different padding, stride, and kernel values for each of the CNN layers. Explain your observations with supporting results obtained from your code.
  - Apply transformations such as translation (shifting the image) and rotation to the input test images. Evaluate how your trained CNN responds to these changes. Based on your observations, explain why CNNs shows robustness to translation but not to rotation. Suggest and experimentally verify one method (architectural or data augmentation) to improve rotation invariance.